

Project Report

Collect all samples into a joint data structure

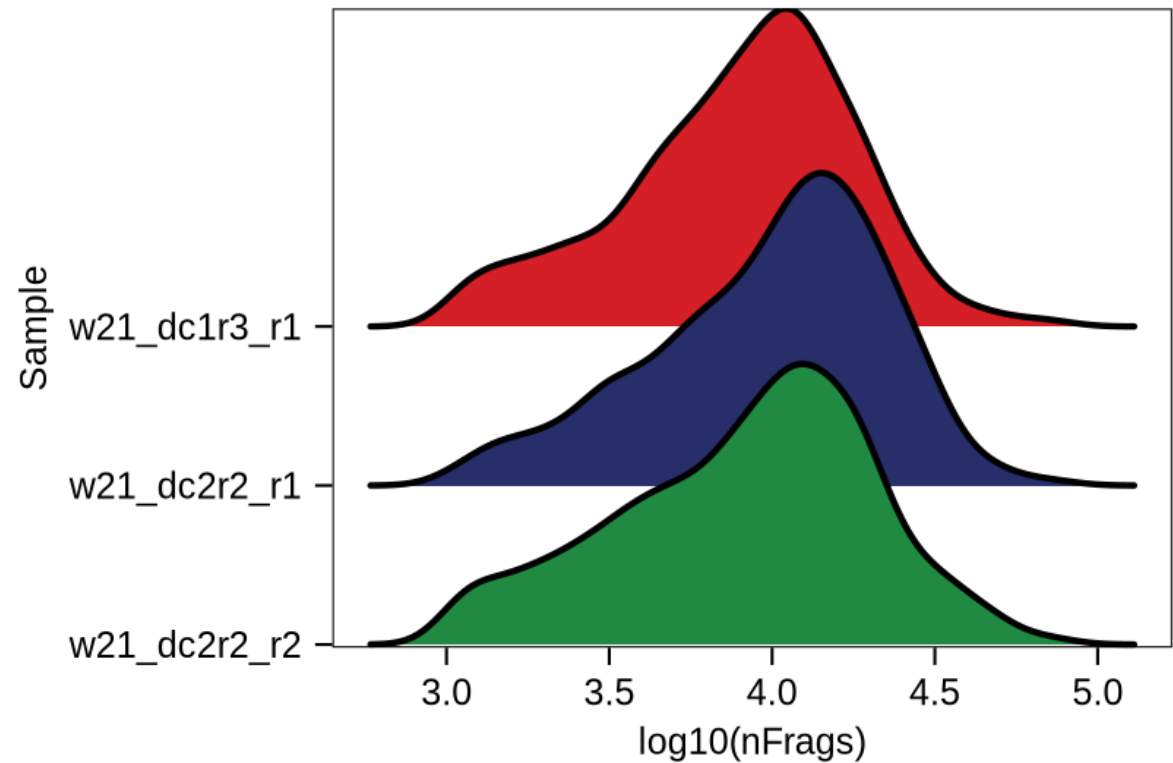
- Number of cells in the project : **9414**
- median TSS-value : **19.878**
- Median of the number of fragments : **9861.5**
- Dimensions of the dataset: **9414 rows and 13 columns**

Quality control

Sample	Number of cells
w21_dc1r3_r1	3319
w21_dc2r2_r1	3621
w21_dc2r2_r2	2474

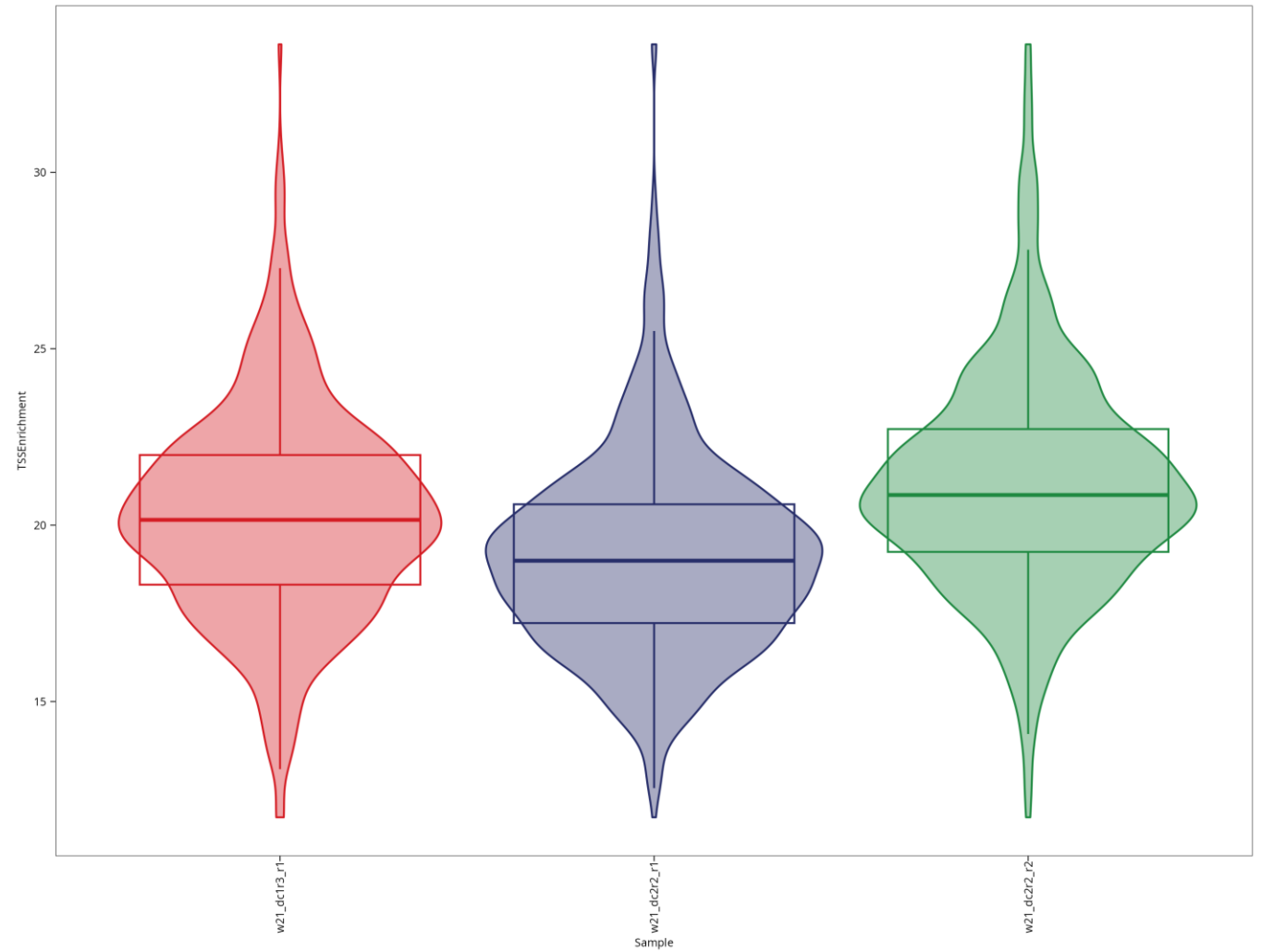
Quality control

the fragment length distribution of all samples in a single plot



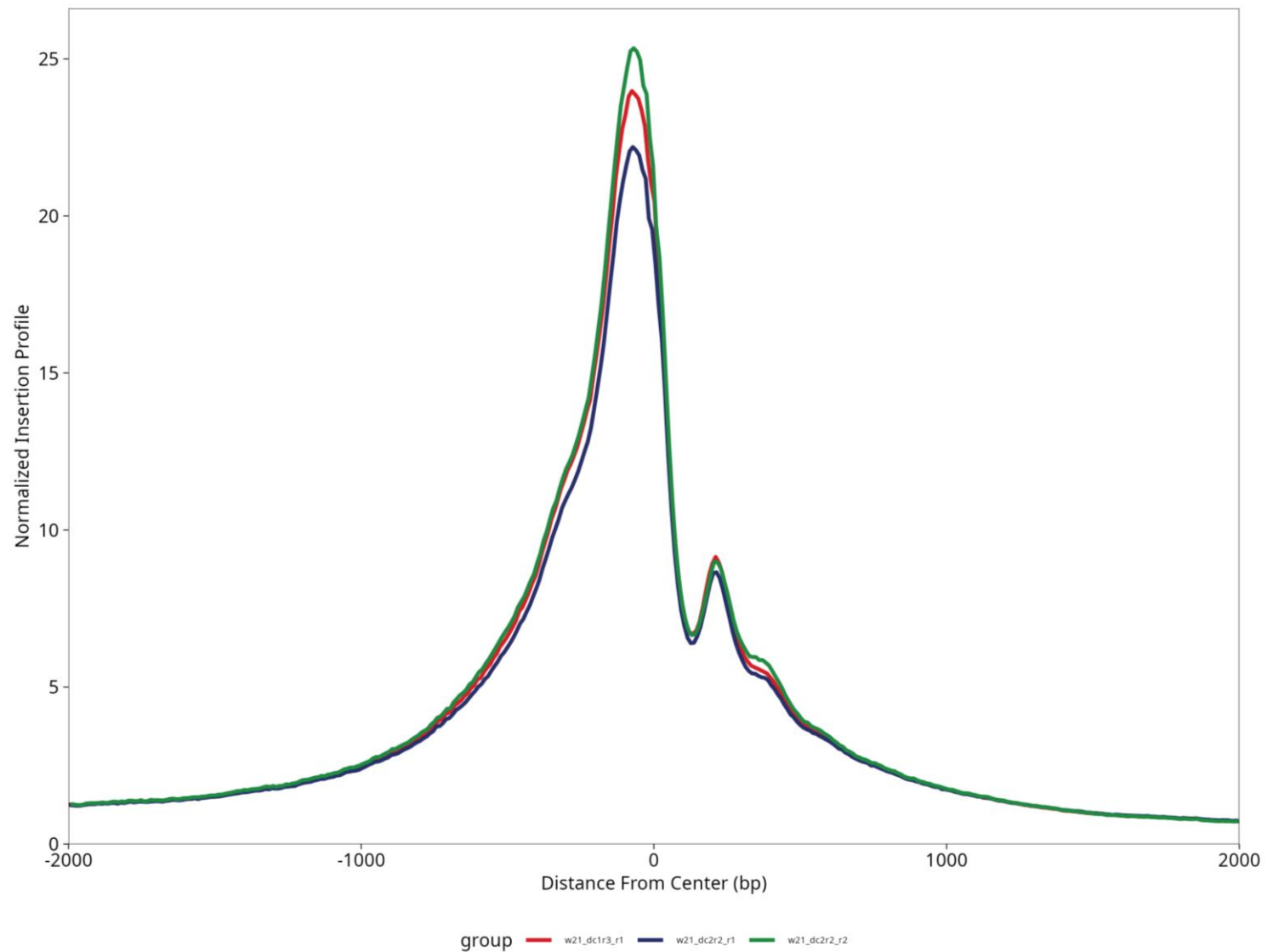
Quality control

- distribution of TSS enrichment scores in each sample



Quality control

distribution of TSS
enrichment scores for all
samples



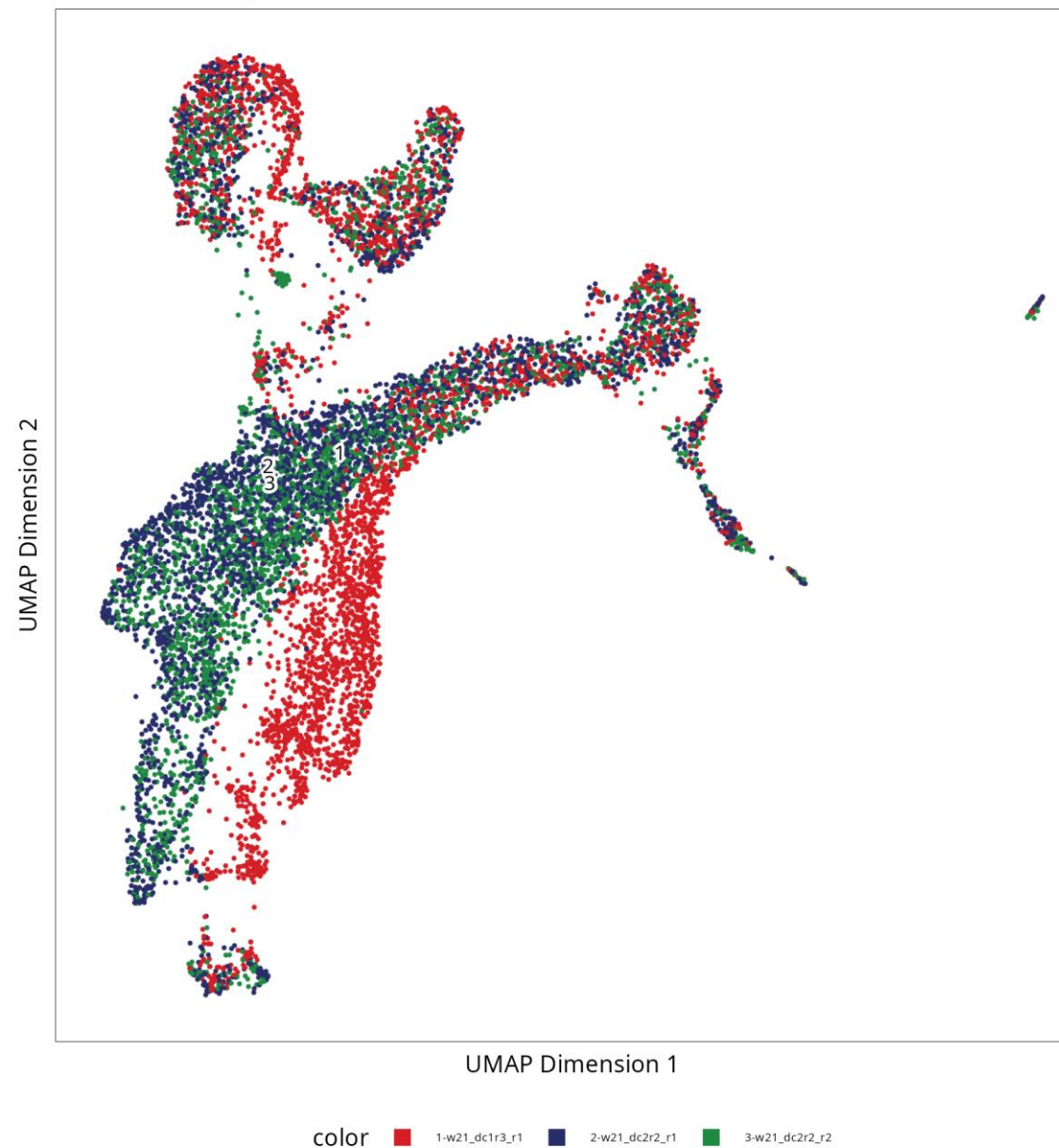
Filter the dataset

- Number of cells in the project after filtration : **9038**
- median TSS-value after filtration : **19.89**
- Median of the number of fragments after filtration : **9546.5**
- Dimensions of the dataset: **9038 rows and 15 columns**

UMAP with sample annotation and QC metrics

- **Sample**

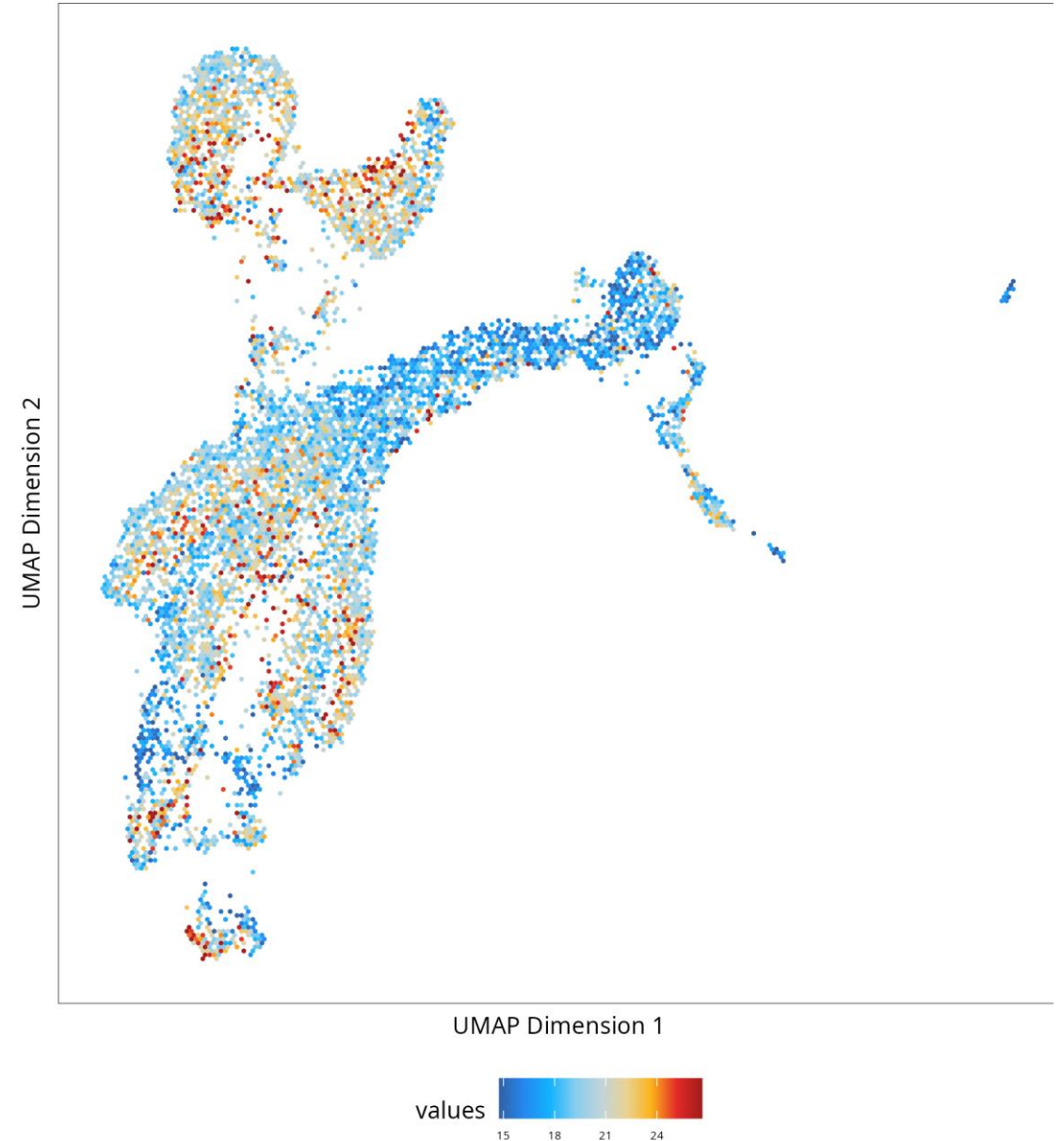
UMAP of IterativeLSI colored by
colData : Sample



UMAP with sample annotation and QC metrics

- **TSSEnrichment**

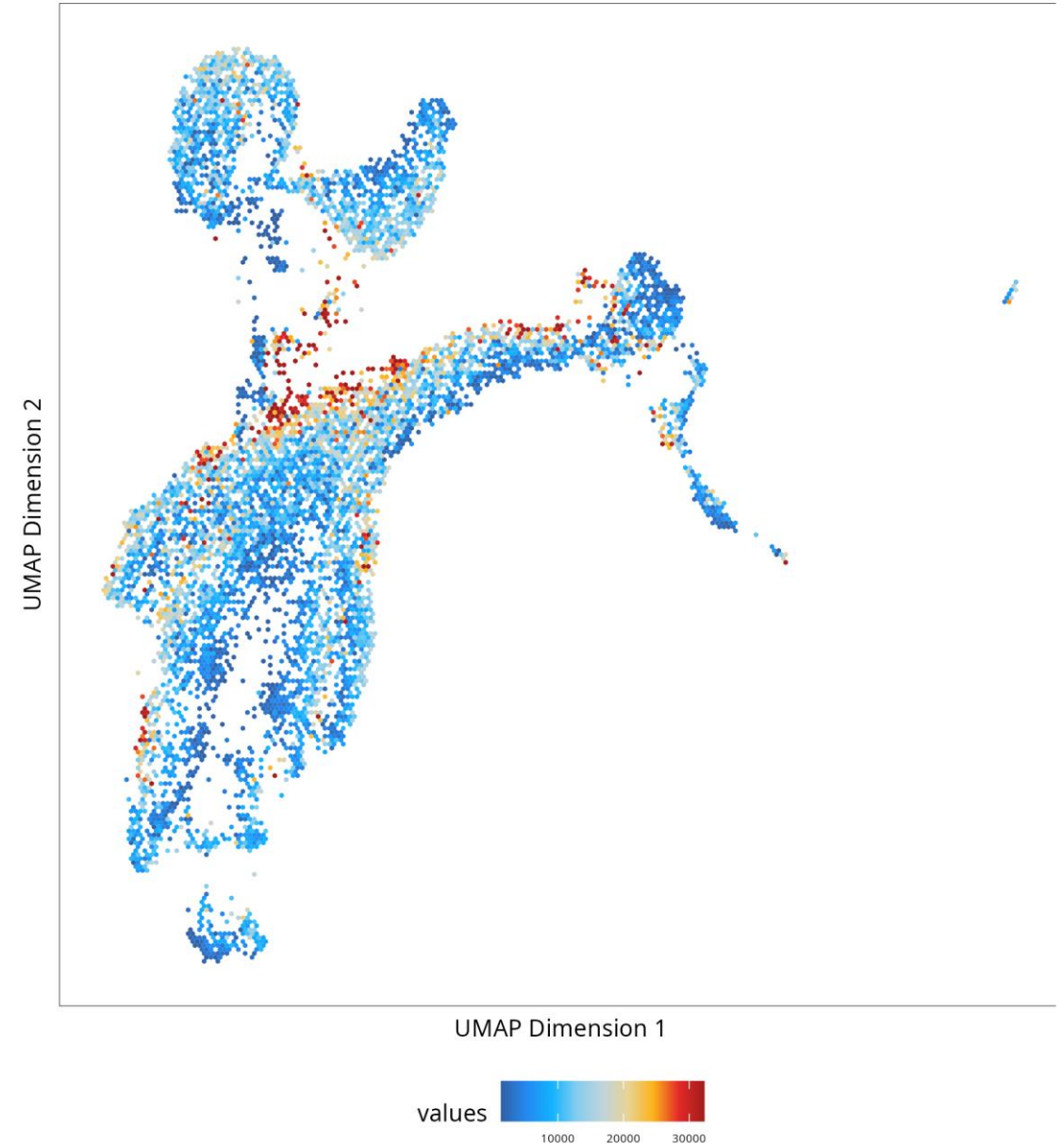
UMAP of IterativeLSI colored by
colData : TSSEnrichment



UMAP with sample annotation and QC metrics

- **nFragments**

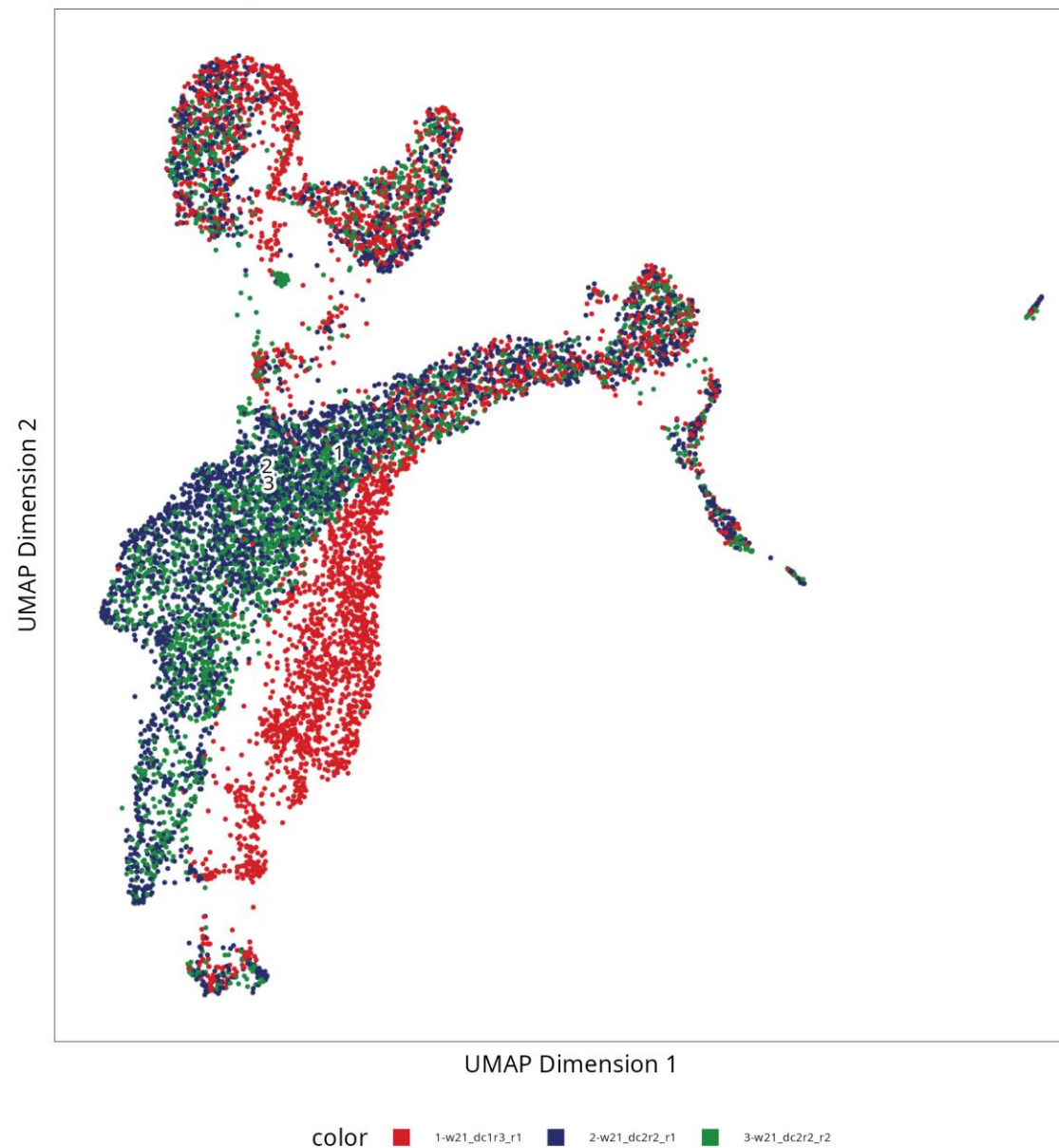
UMAP of IterativeLSI colored by
colData : nFrag



Dealing with batch effects

- **Before Batch correction**

UMAP of IterativeLSI colored by
colData : Sample

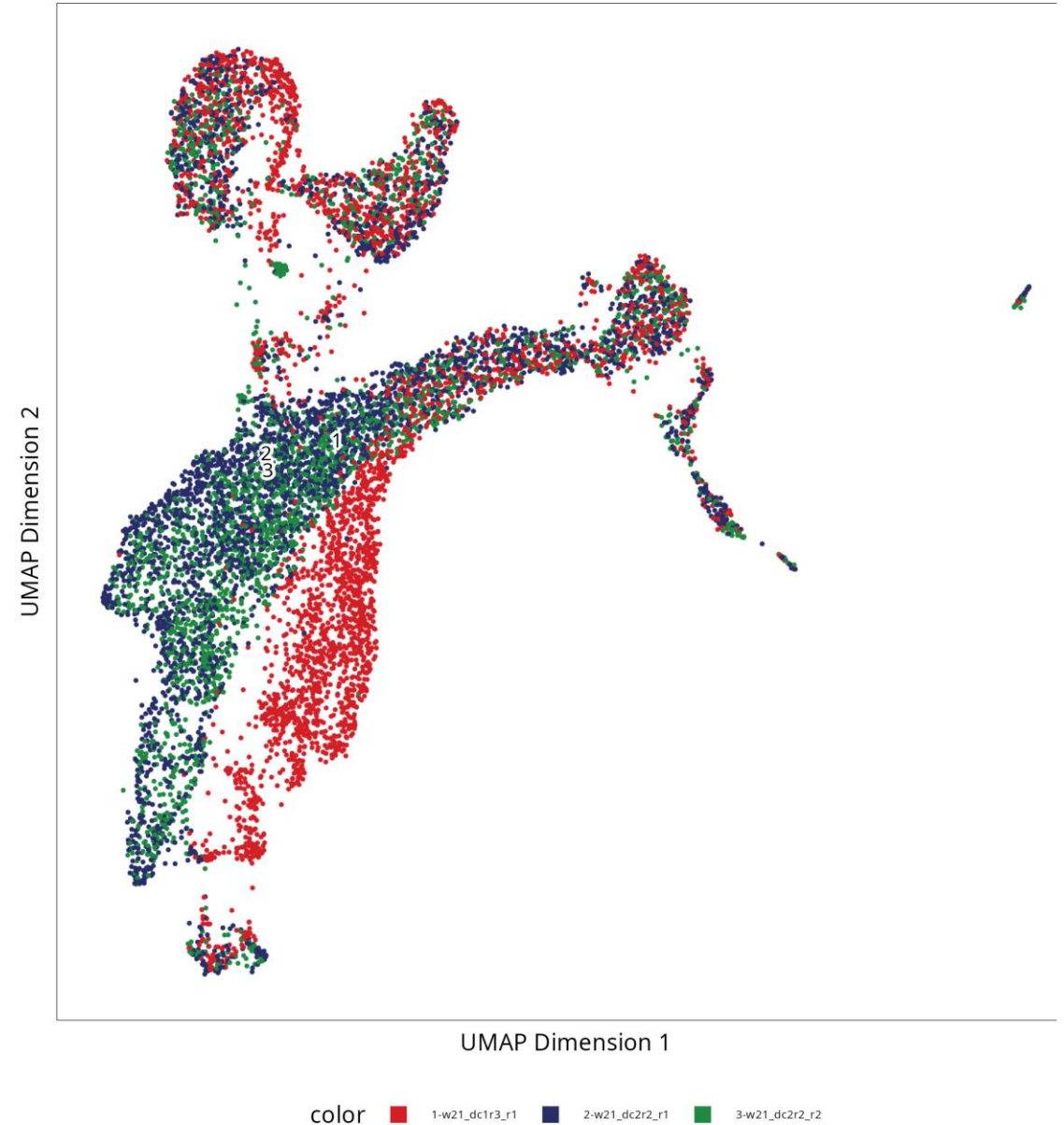


Dealing with batch effects

- **After Batch correction**

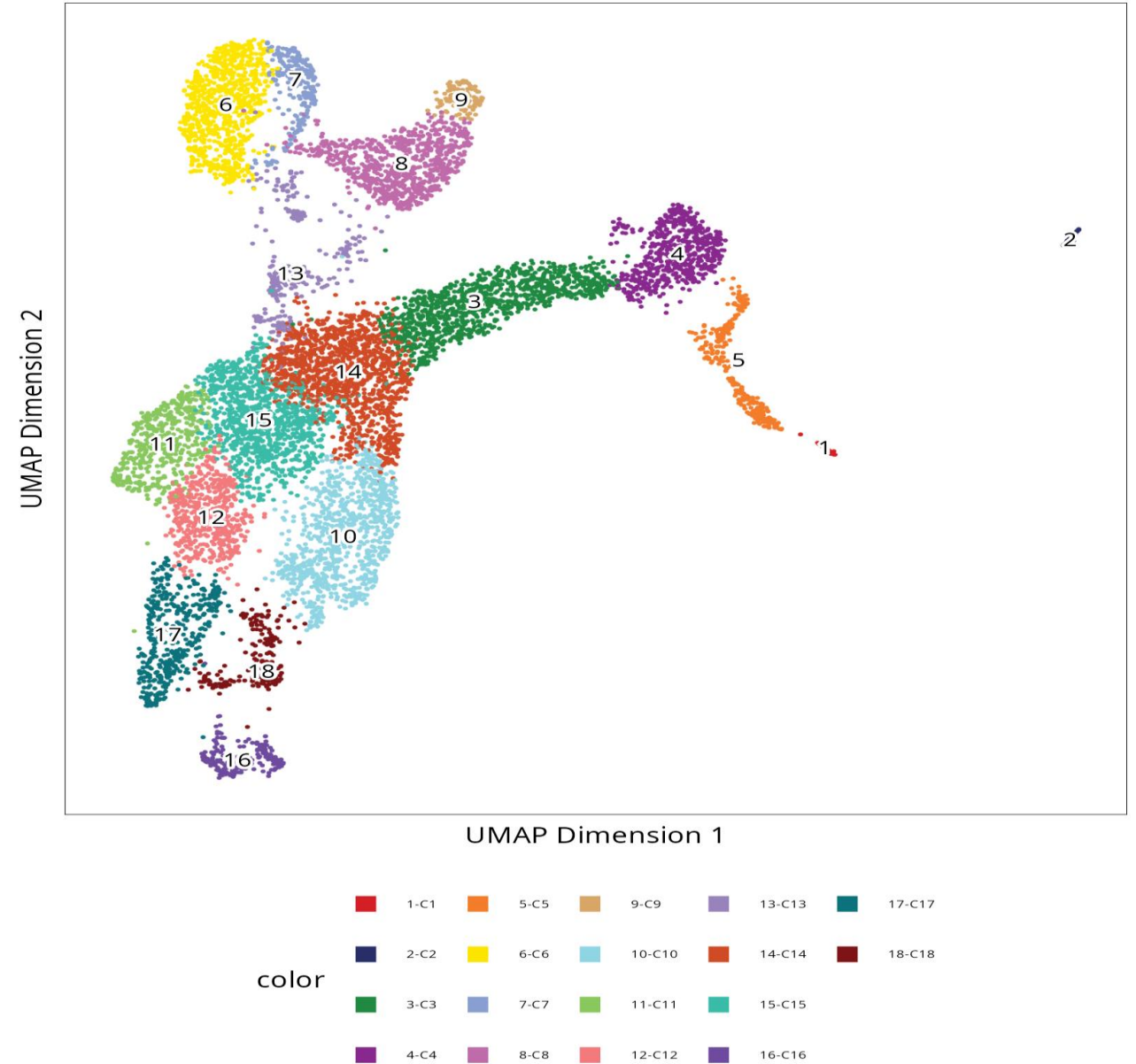
There is no batch effect between cells in the different samples

UMAP of IterativeLSI colored by
colData : Sample



Clustering

UMAP of IterativeLSI colored by
colData : Clusters



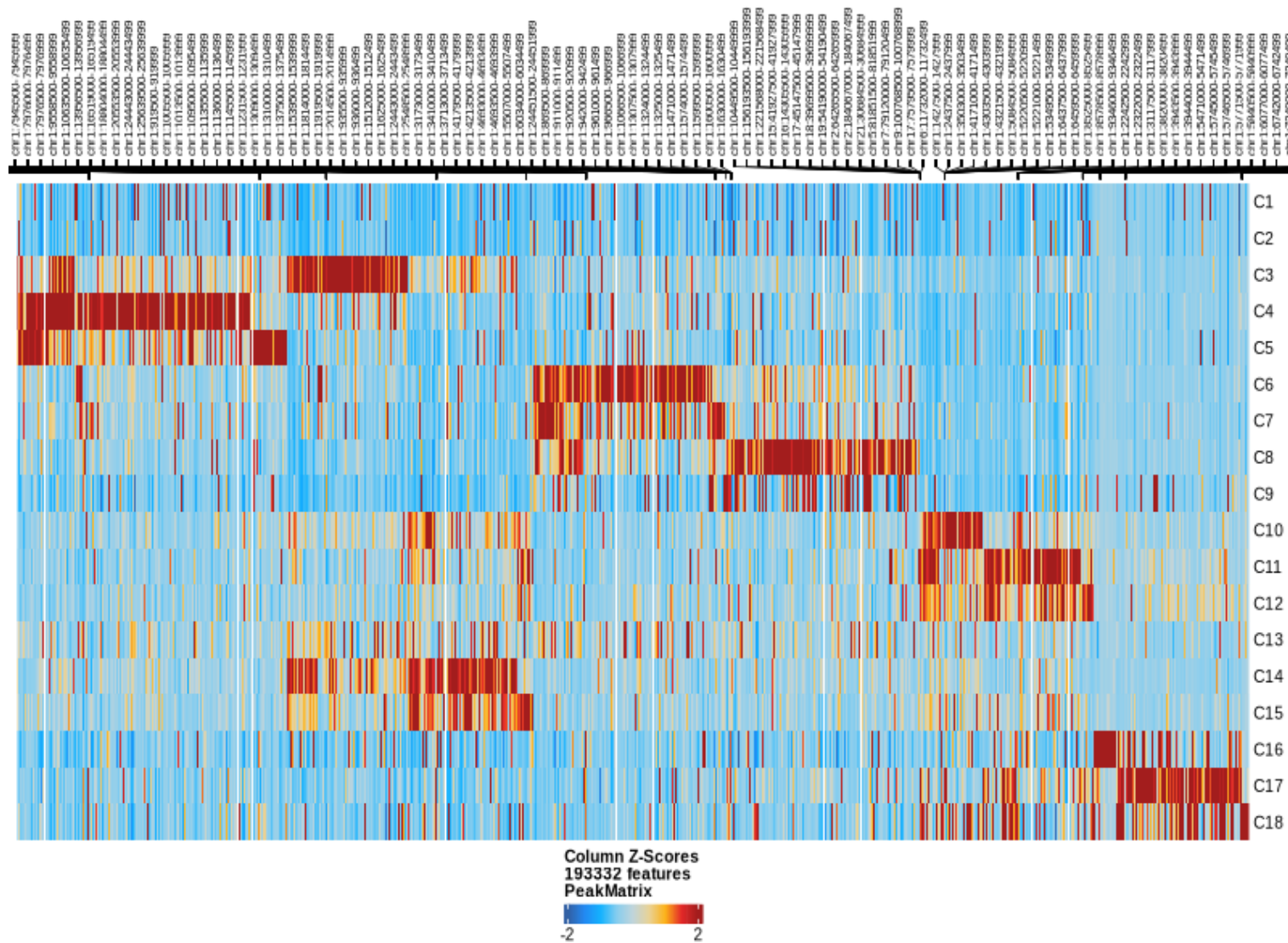
Clustering



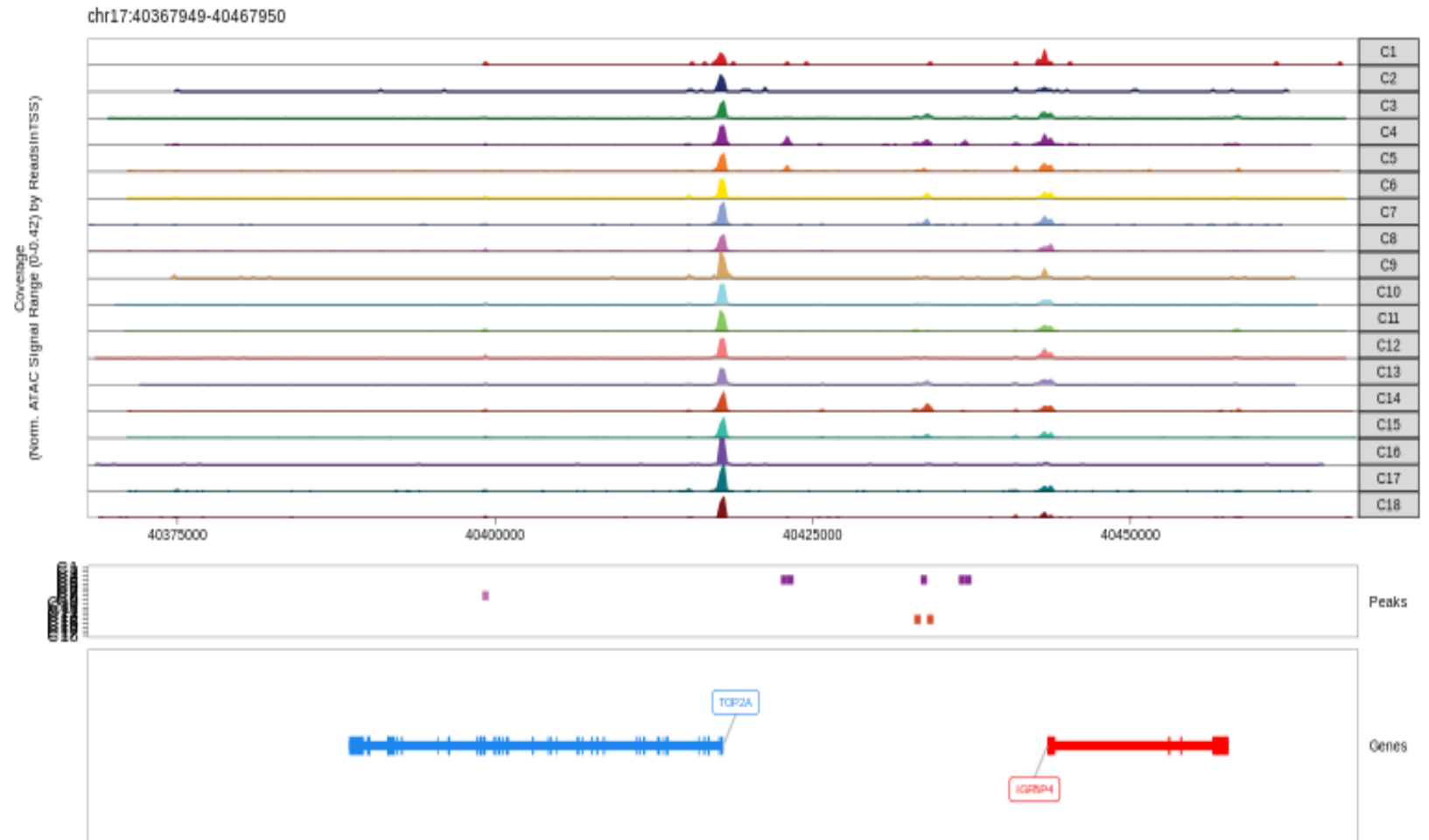
Sample proportion per cluster

```
18 x 3 sparse Matrix of class "dgCMatrix"
  w21_dc2r2_r1 w21_dc1r3_r1 w21_dc2r2_r2
C14          551          384          348
C3           430          306          268
C1            12           6           9
C13           70          121           82
C15          535           57          416
C4           217          220          155
C17          228           6          176
C11          307           1          151
C6           293          308          184
C16           83           60           43
C8           252          334          176
C12          266           7          254
C5           122           82           76
C2            26           10           12
C7            19          204           14
C9            27           52           23
C18           14          171           7
C10            1          860           2
```


Heatmap showing accessibility in the marker peaks



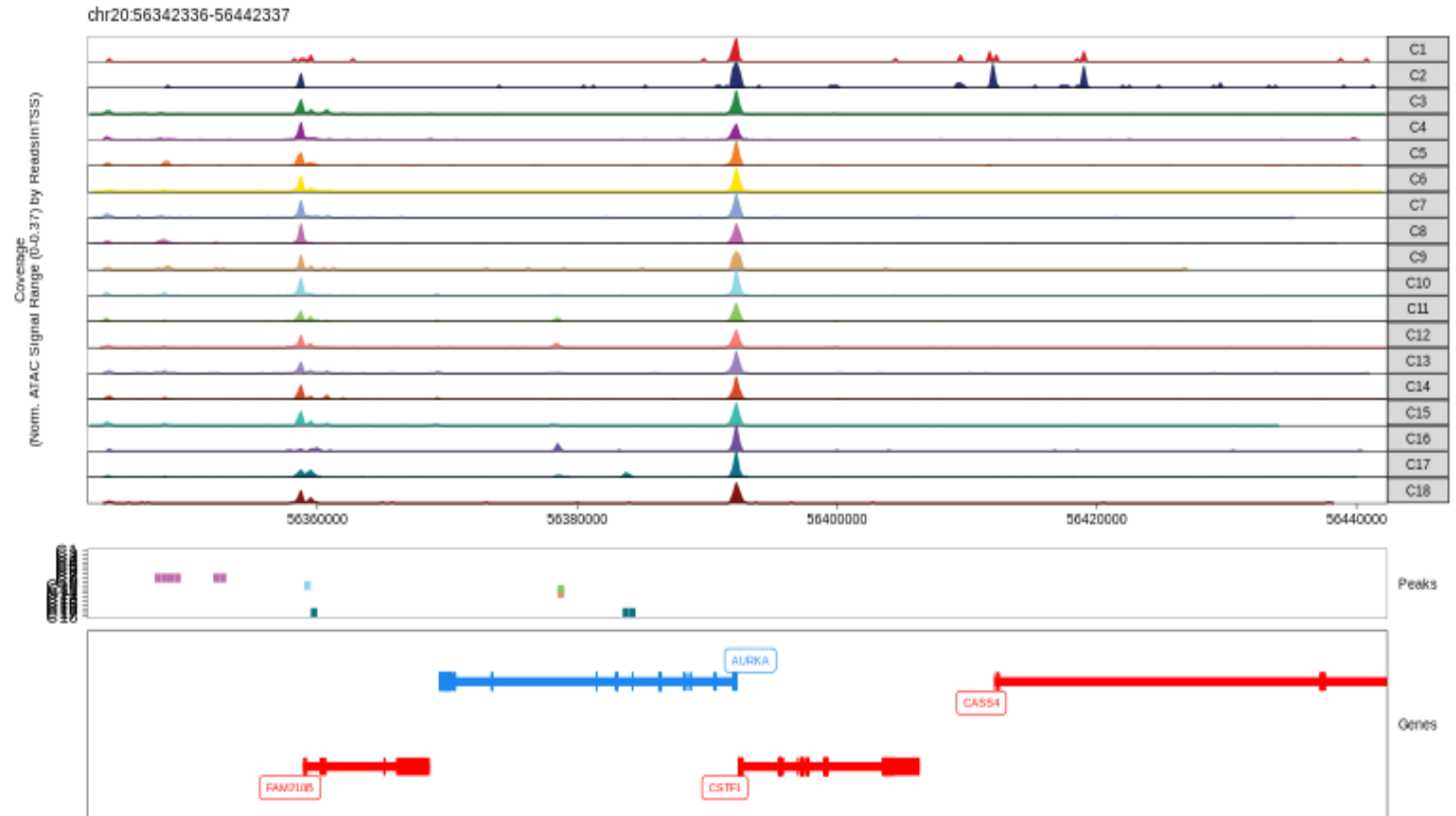
- browser plot for TOP2A



- browser plot for MKI67



- browser plot for AURKA



- 

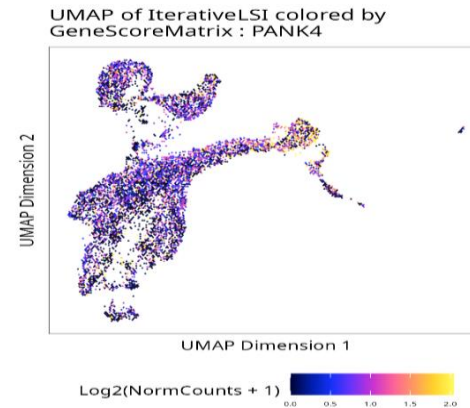
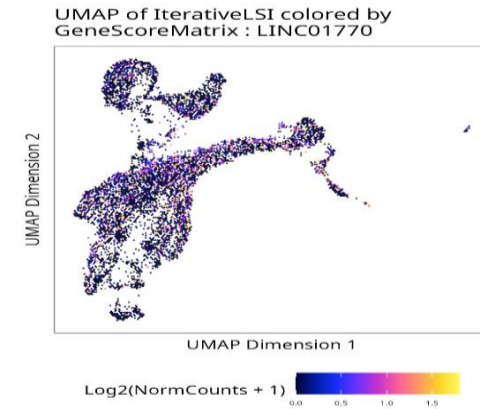
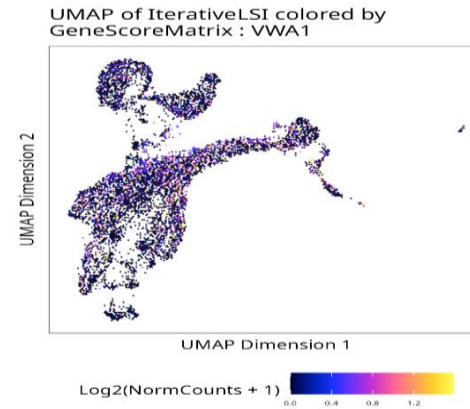
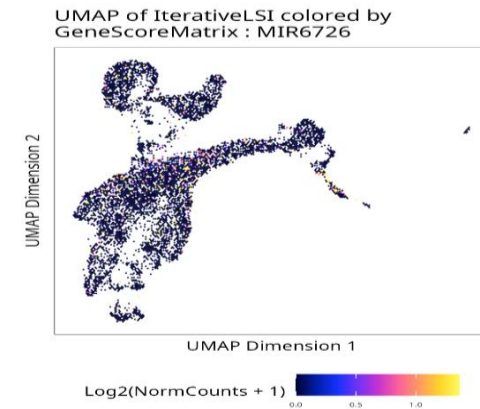
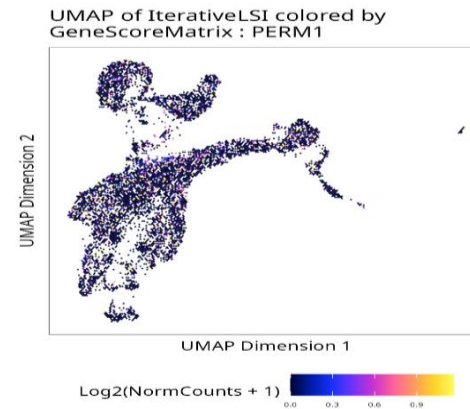


- browser plot for SLC12A7



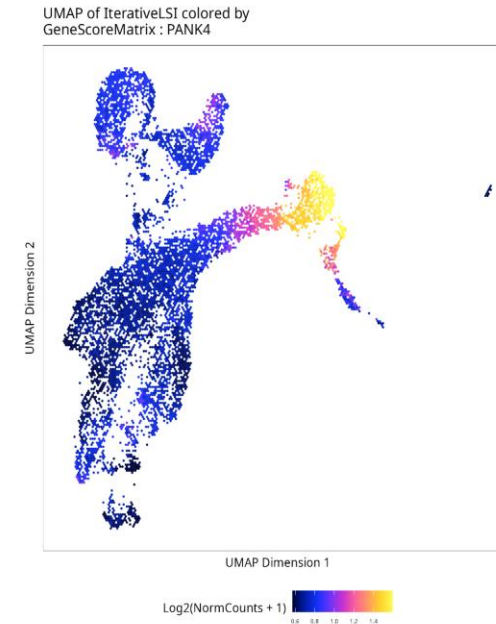
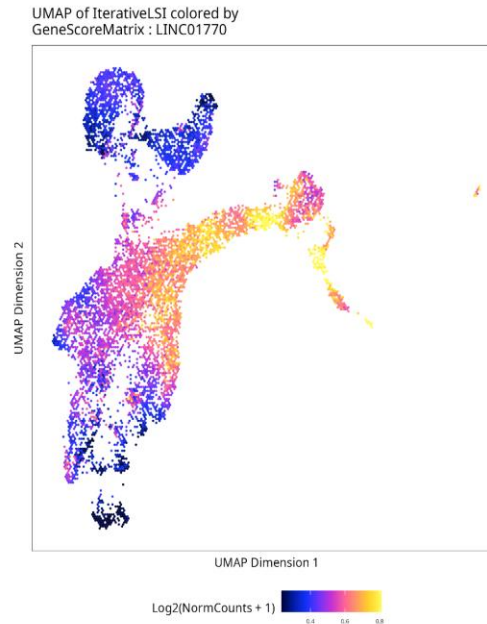
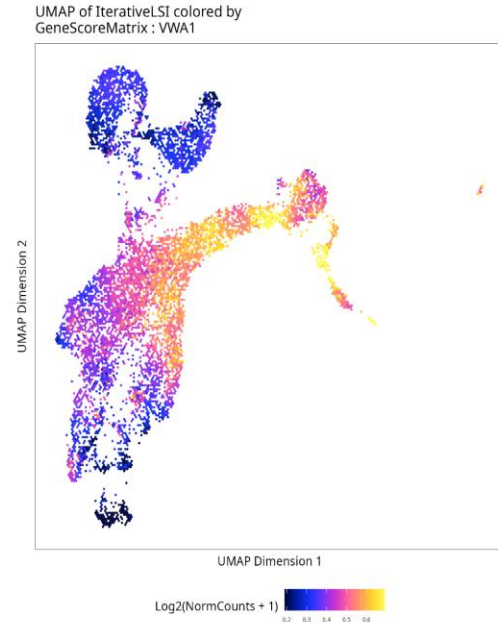
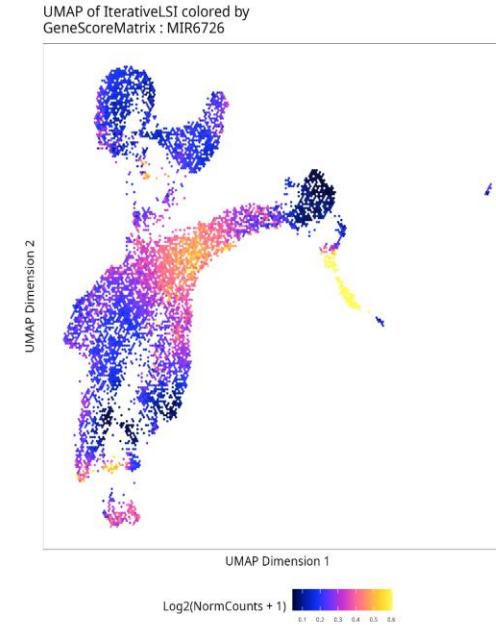
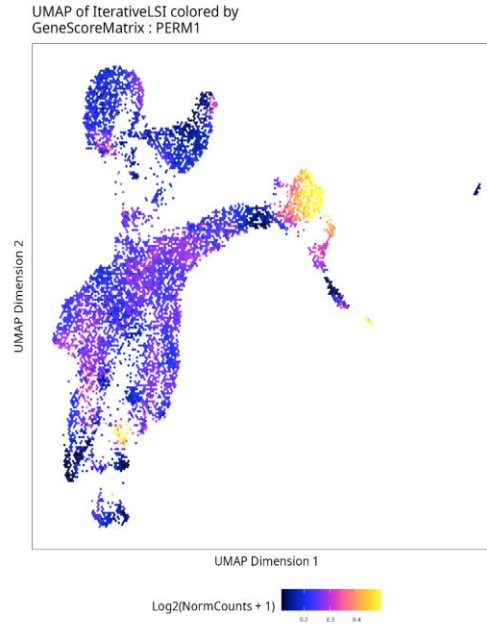
Using MAGIC

- Without MAGIC



Using MAGIC

- **With MAGIC:**
- Markov Affinity-based Graph Imputation of Cells (MAGIC) is an algorithm for denoising high-dimensional data most commonly applied to single-cell RNA sequencing data. MAGIC learns the manifold data, using the resultant graph to smooth the features and restore the structure of the data.

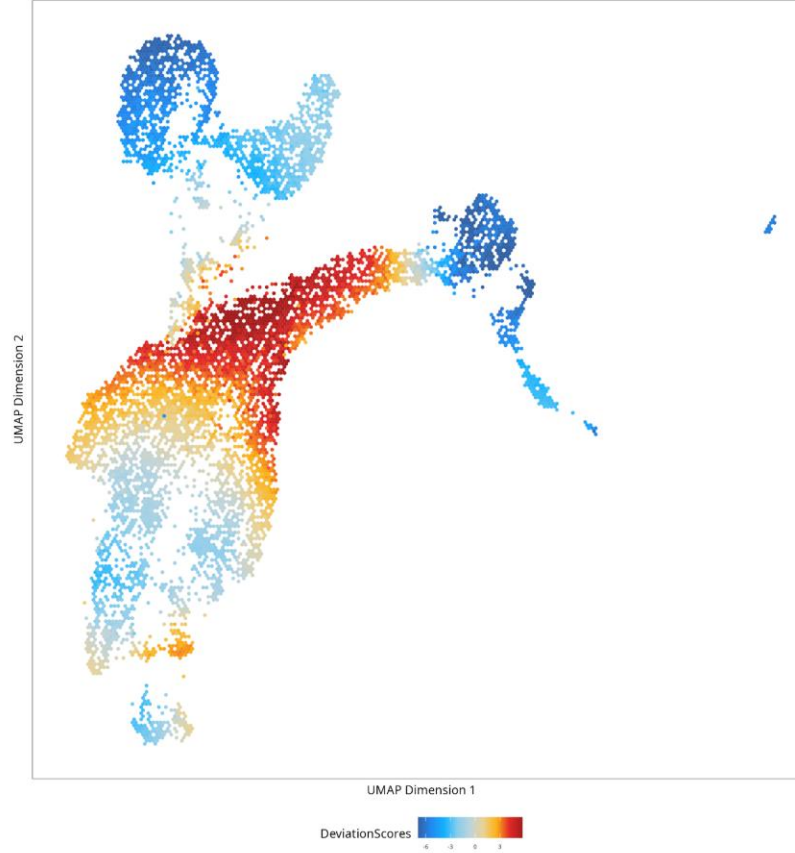


Compute TF motif activity

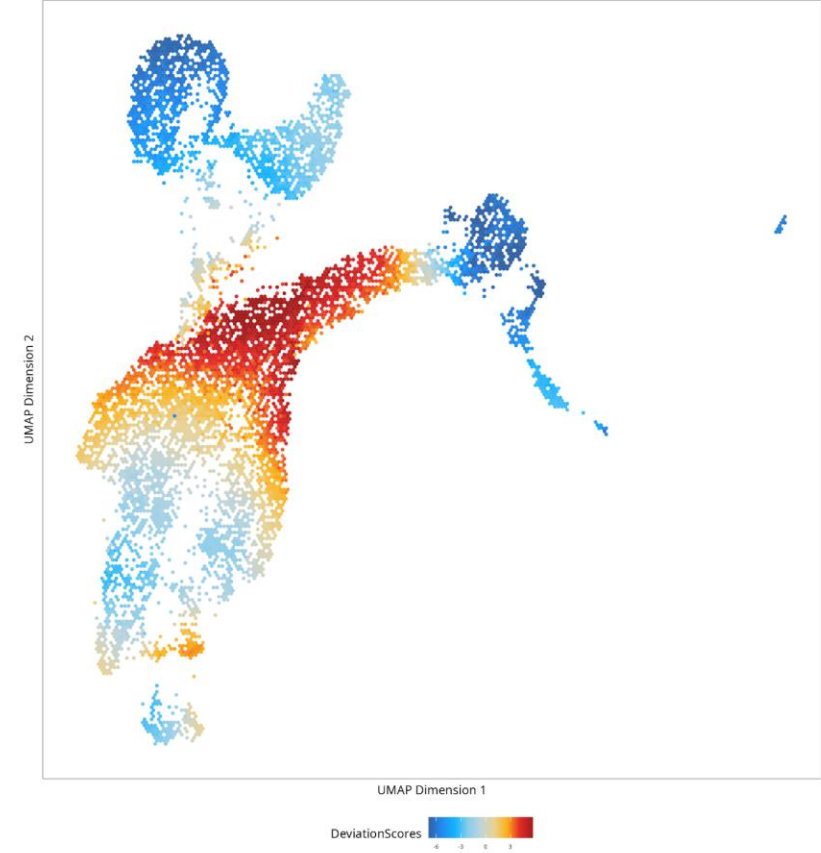
- We used the **cisbp** annotation dataset
- This set was obtained chromVAR package to give the motifs that are contained in peaks

Plot UMAP embeddings for marker TFs

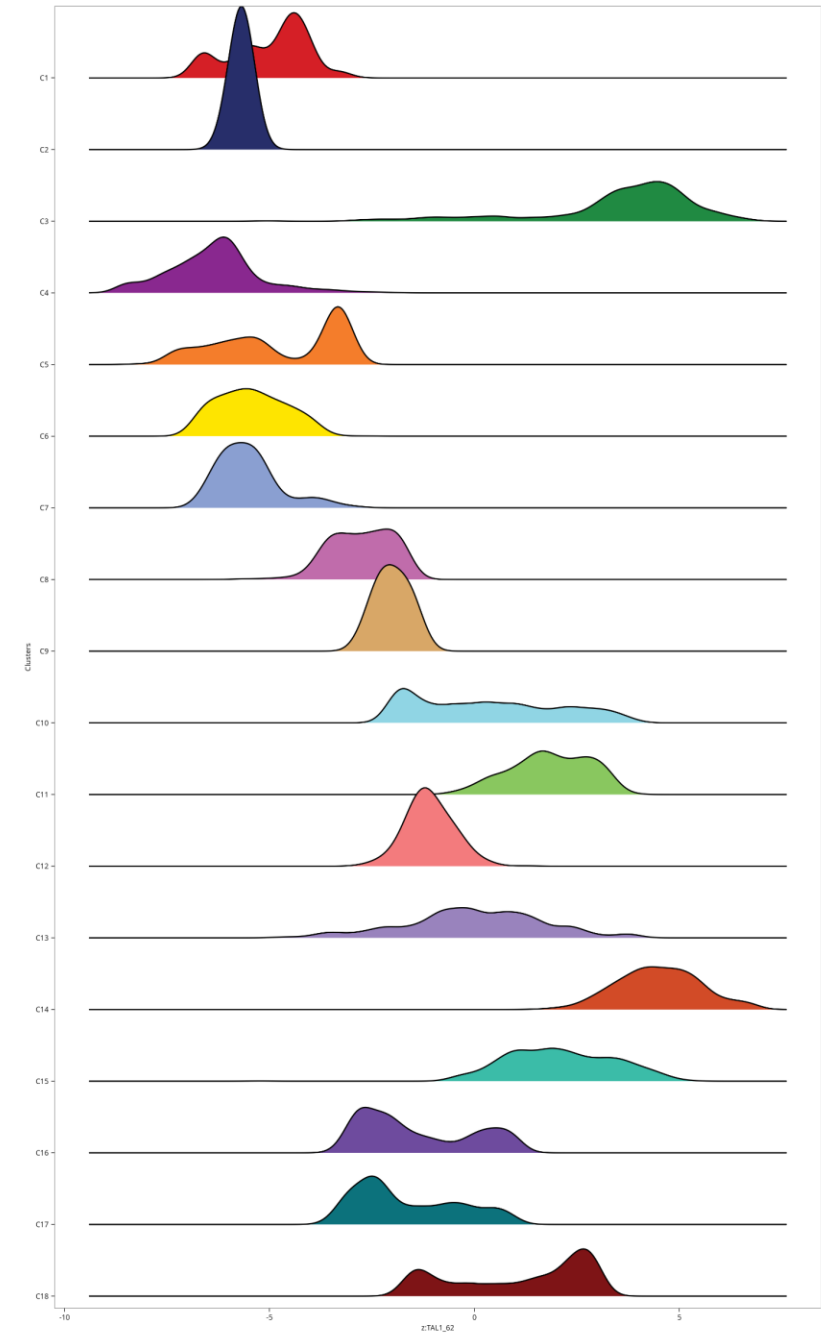
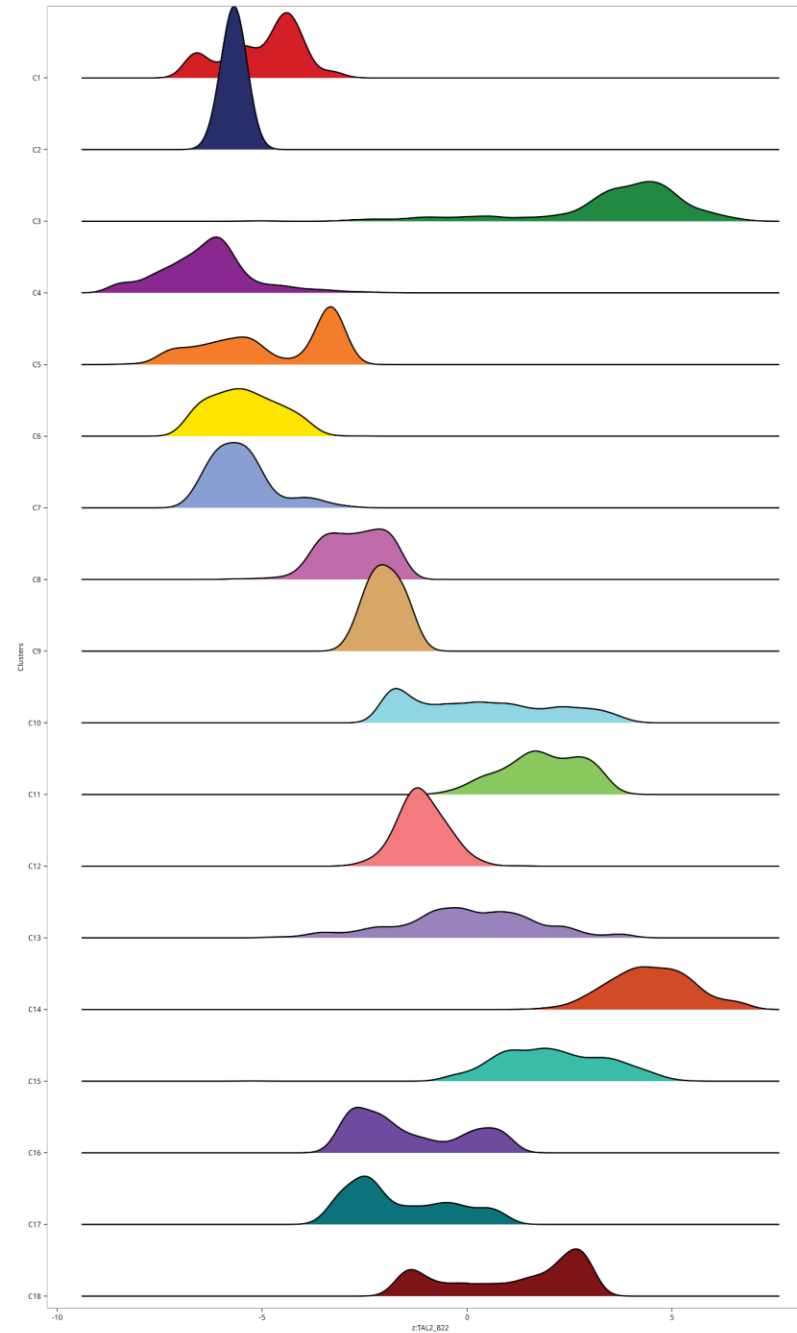
UMAP of IterativeLSI colored by
MotifMatrix : z:TAL2_822



UMAP of IterativeLSI colored by
MotifMatrix : z:TAL1_62

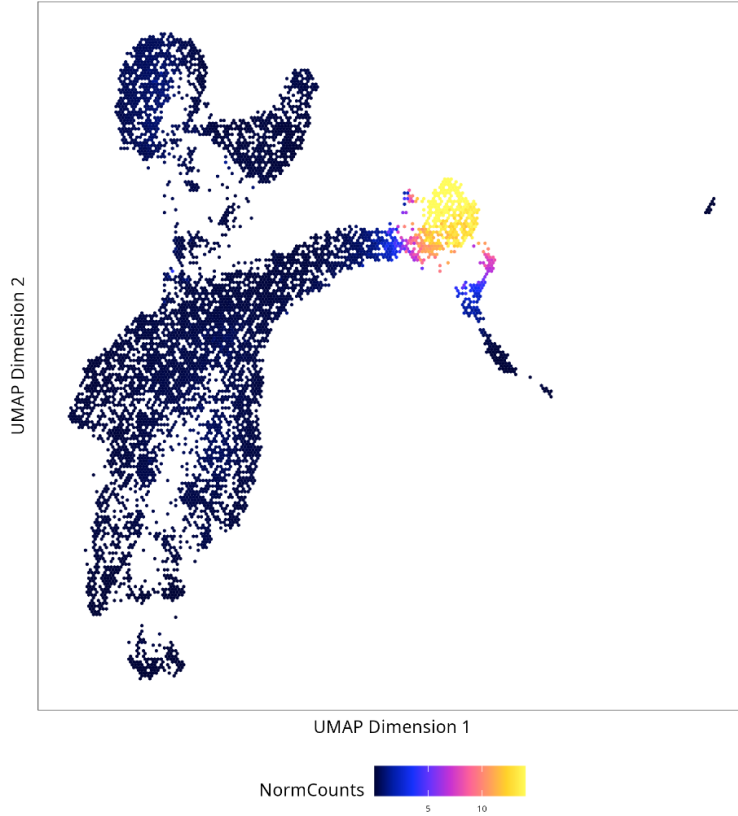


THE distribution
PLOT of motif
activity of marker
TFs in the different
clusters

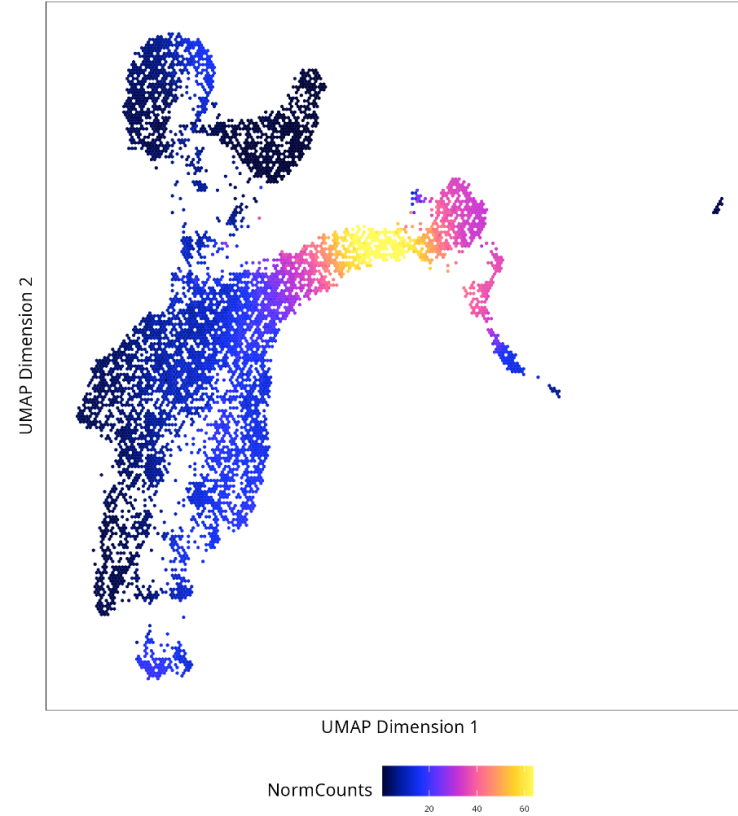


Data integration

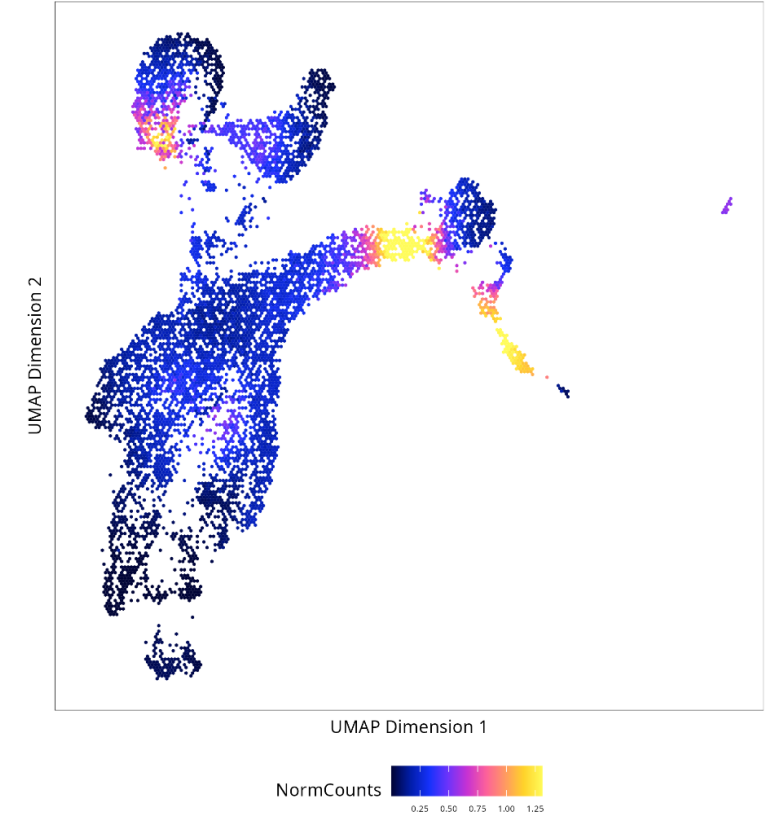
UMAP of IterativeLSI colored by
GeneIntegrationMatrix : ID4



UMAP of IterativeLSI colored by
GeneIntegrationMatrix : NFIA

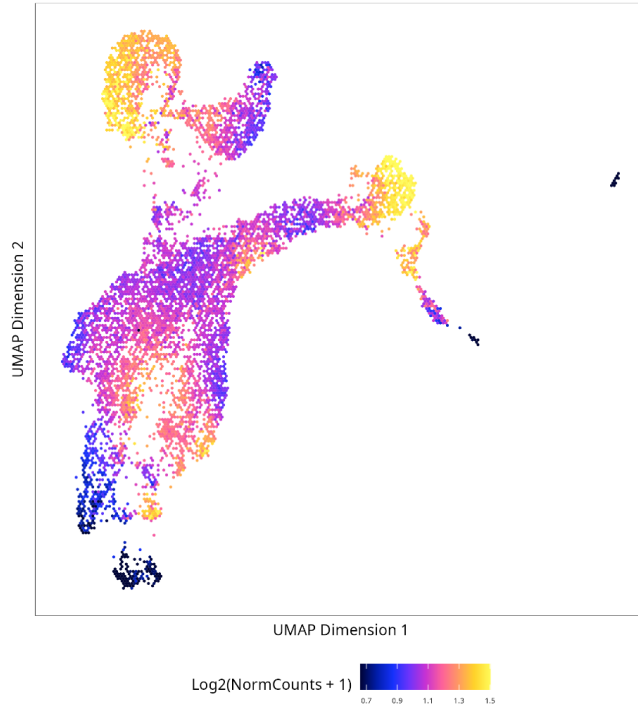


UMAP of IterativeLSI colored by
GeneIntegrationMatrix : ASCL1

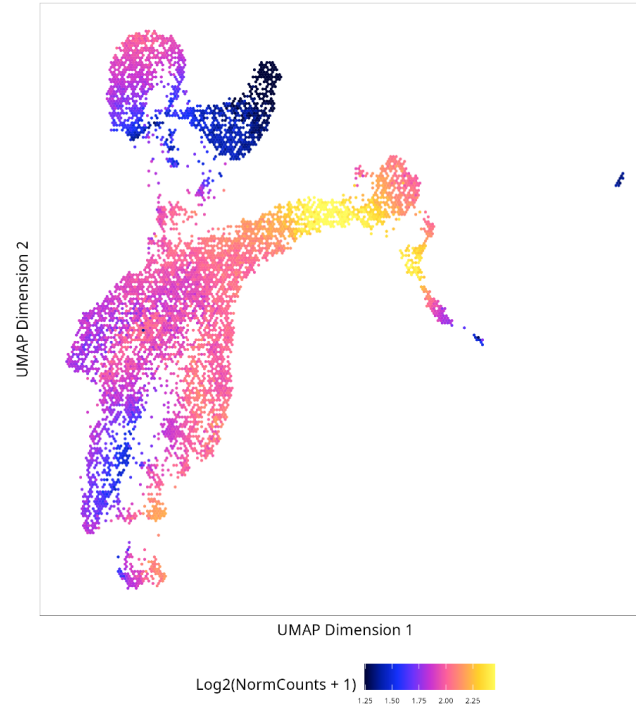


UMAP space and overlay it with gene expression levels for the highlighted marker genes in table 2

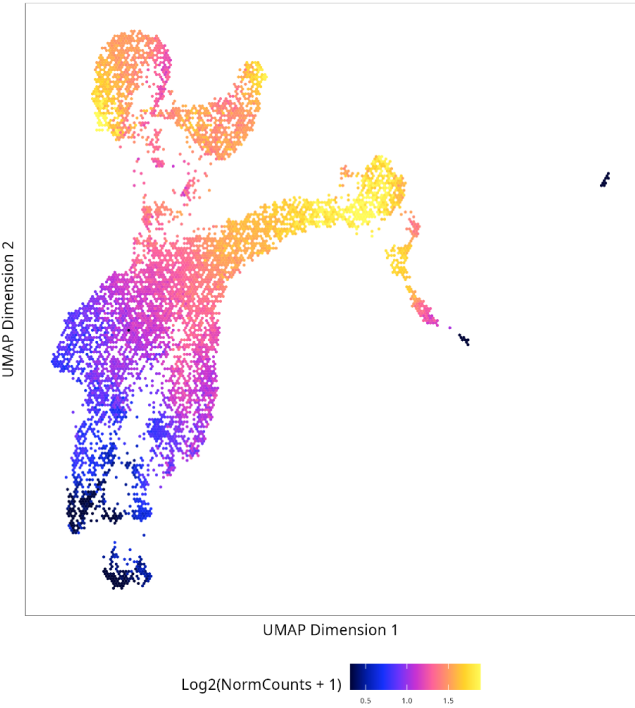
UMAP of IterativeLSI colored by
GeneScoreMatrix : ID4



UMAP of IterativeLSI colored by
GeneScoreMatrix : NFIA



UMAP of IterativeLSI colored by
GeneScoreMatrix : ASCL1



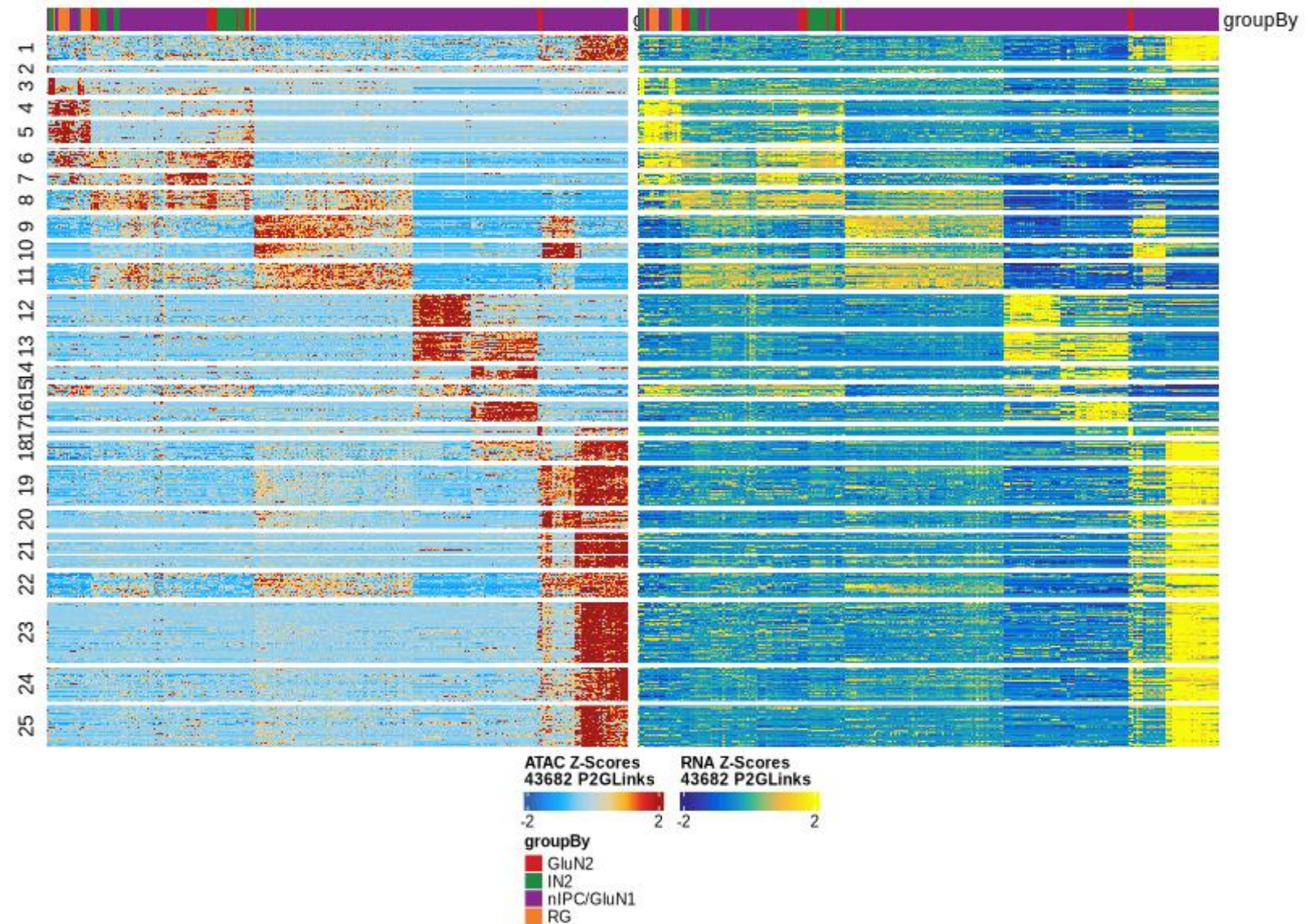
Correlation Coefficients

The gene exhibit the highest.

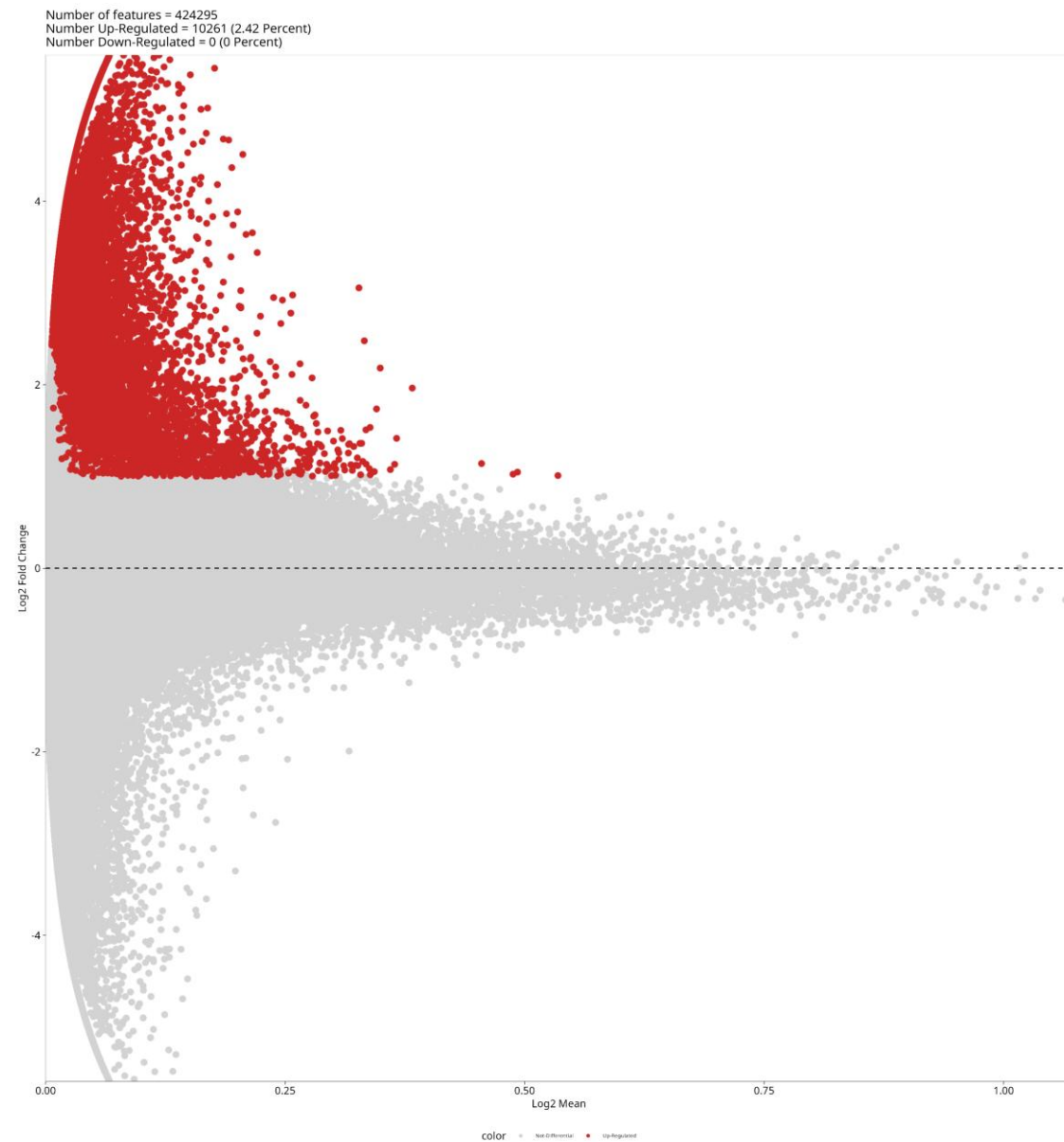
```
> head(order_vale,4)
DataFrame with 4 rows and 17 columns
  GeneScoreMatrix_name GeneIntegrationMatrix_name
      <character>          <character>
1             A1BG             A1BG
2             A1BG             A1BG-AS1
3          A1BG-AS1             A1BG
4          A1BG-AS1             A1BG-AS1
```

```
> tail(order_vale,4)
DataFrame with 4 rows and 17 columns
  GeneScoreMatrix_name GeneIntegrationMatrix_name
      <character>          <character>
[20730,]          ZYG11B          ZYG11B
[20731,]           ZYX          ZYX
[20732,]         ZZEF1         ZZEF1
[20733,]          ZZZ3          ZZZ3
```

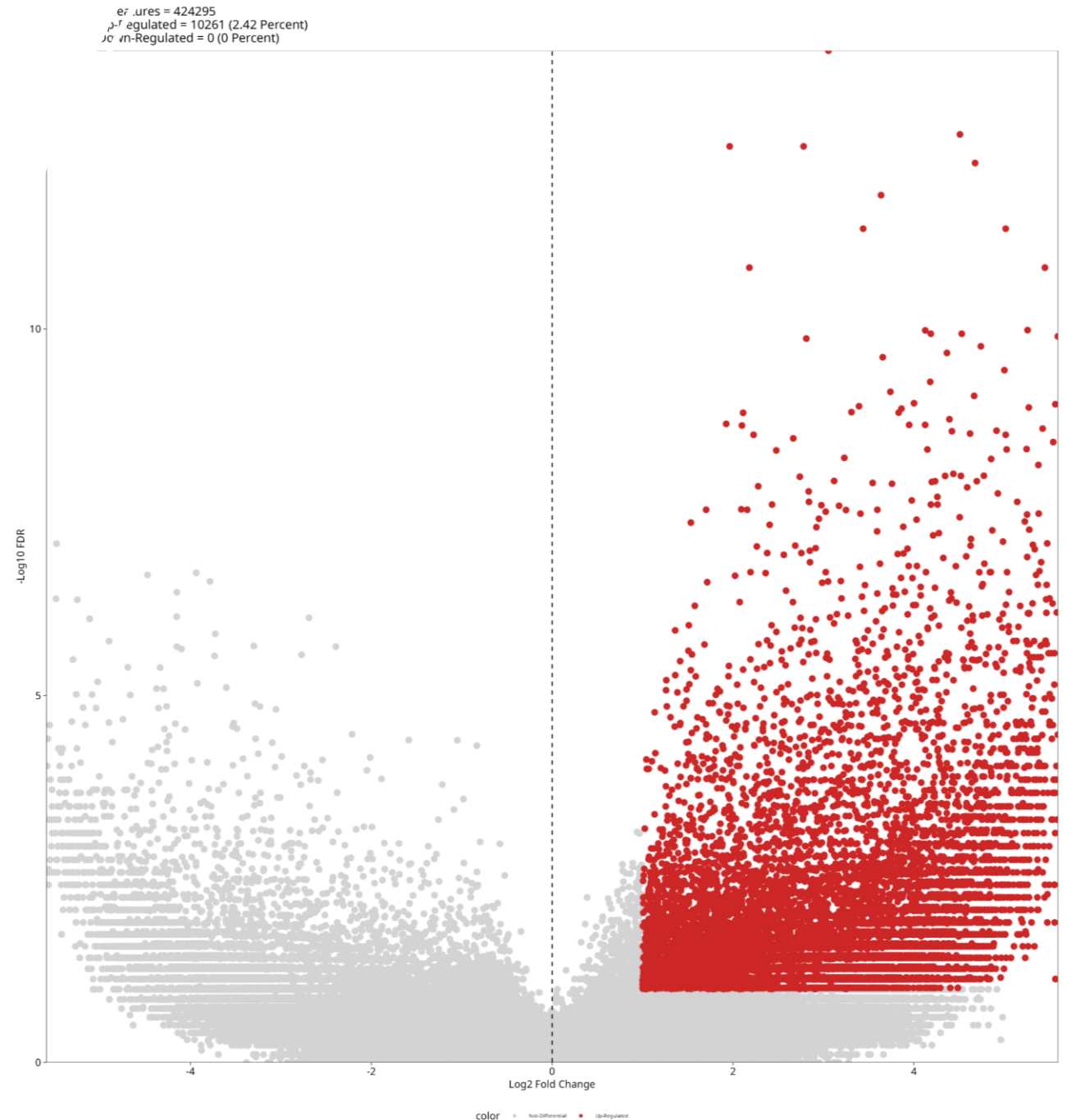

side-by-side heatmaps



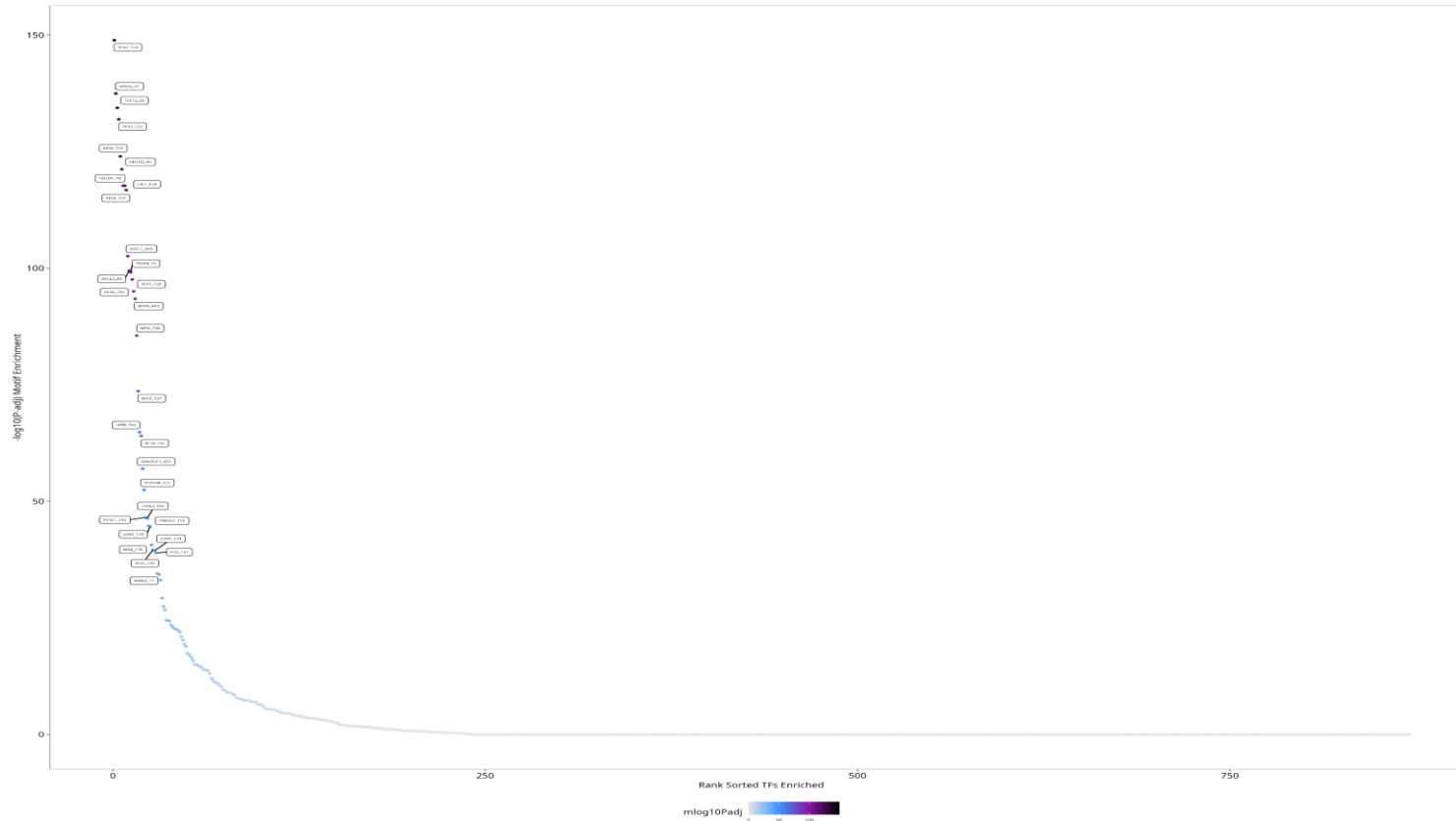
Differential peak accessibility (MA)



Differential peak accessibility (Volcano)

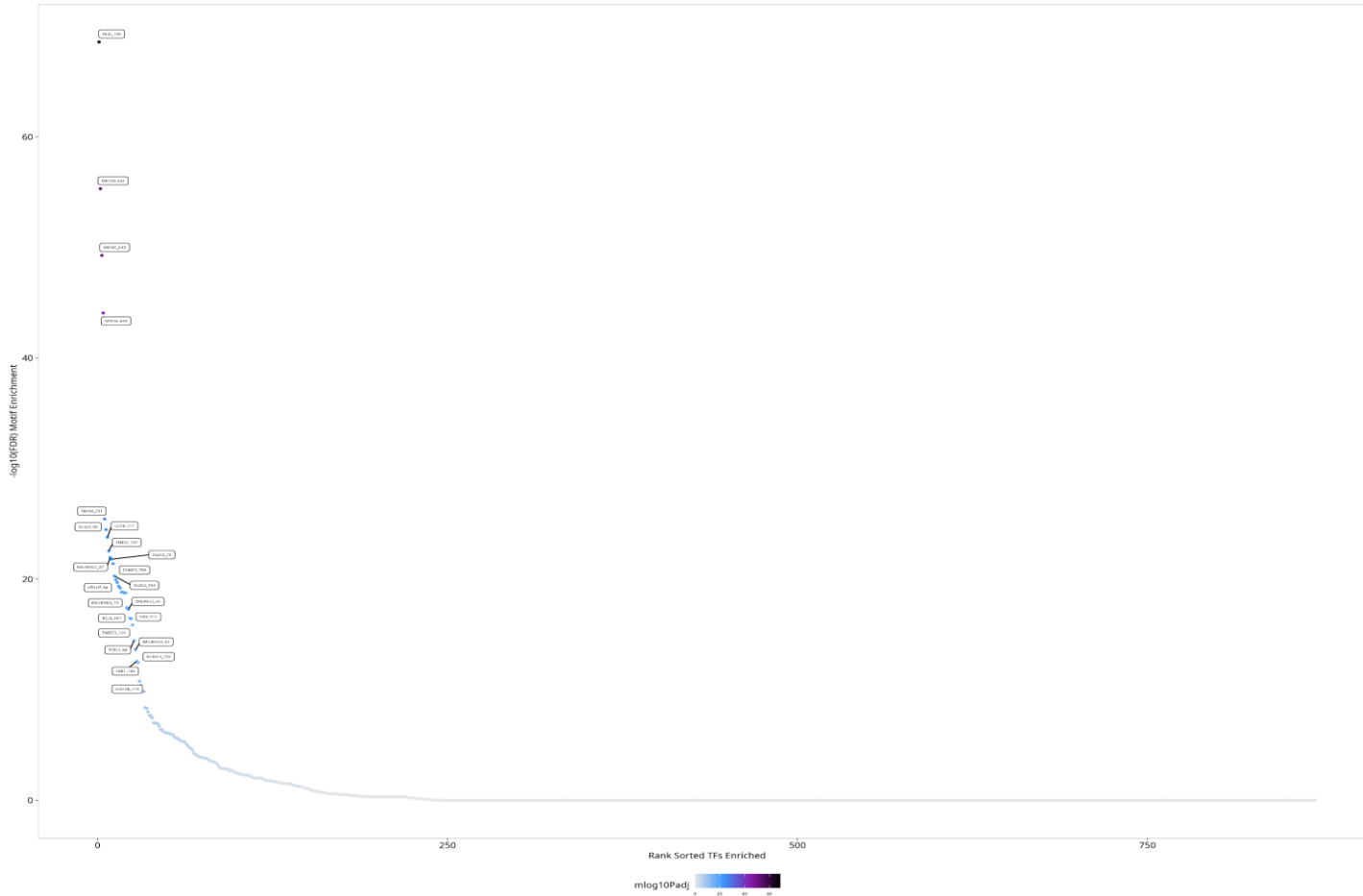


TF motif enrichment



```
> head(df)
      TF mlog10Padj rank
724 RFX2_724    148.9126   1
 41  MYOG_41    137.4762   2
 56  TCF12_56    134.4136   3
723 RFX3_723    131.9617   4
729 RFX6_729    123.9989   5
 80  NHLH2_80    121.2367   6
```

TF motif enrichment



```
> head(dfDo)
      TF mlog10Padj rank
740  NFIC_740    68.58866   1
642  MEF2D_642    55.31146   2
640  MEF2C_640    49.28366   3
639  MEF2A_639    44.07406   4
791  TBX10_791    25.44546   5
90   OLIG1_90     24.48156   6
```

TF motif enrichment

